

Router basic configuration:

This section includes IOS commands that are absolutely identical on both routers and switches, except the part of *line aux 0* which is configured only on router because switches do not have an auxiliary port.

```
1
2
3 Router(config)# hostname R1
4 R1(config)# enable secret cisco
5 R1(config)# line con 0
6 R1(config-line)# password cisco
7 R1(config-line)# login
8 R1(config-line)# logging synchronous
9 R1(config-line)# exec-timeout 30 0
10 R1(config-line)# exit
11 R1(config)# line vty 0 4
12 R1(config-line)# password cisco
13 R1(config-line)# login
14 R1(config-line)# logging synchronous
15 R1(config-line)# exec-timeout 30 0
16 R1(config-line)# exit
17 R1(config)# line aux 0
18 R1(config-line)# password cisco
19 R1(config-line)# login
20 R1(config-line)# logging synchronous
21 R1(config-line)# exec-timeout 30 0
22 R1(config-line)# exit
23 R1(config)# banner motd $
24 -----
25 UNAUTHORIZED ACCESS IS PROHIBITED
26 -----
27 $
28 R1(config)# alias exec c configure terminal
29 R1(config)# alias exec s show ip interface brief
30 R1(config)# alias exec sr show running-config
31 R1(config)# no ip domain-lookup
32 R1(config)# service password-encryption
33 R1(config)# ip domain-name example.com
34 R1(config)# username admin password cisco
35 R1(config)# crypto key generate rsa
36 How many bits in the modulus [512]: 1024
37 R1(config)# ip ssh version 2
38 R1(config)# line vty 0 4
39 R1(config-line)# login local
40 R1(config-line)# transport input telnet ssh
41
42
43
```

Configuring router interfaces:

Clock rate is set only on the DCE side, typically the ISP side. On your router which is DTE you don't need to set clocking.

```
1
2 R1(config)# interface fastEthernet 0/0
3 R1(config-if)# description LINK TO LOCAL LAN THROUGH SW1
```

```

3 R1(config-if)# ip address 172.16.1.1 255.255.255.0
4 R1(config-if)# no shutdown
5 R1(config-if)# exit
6 R1(config)# interface serial 0/1/0
7 R1(config-if)# description WAN CONNECTION TO R2
8 R1(config-if)# ip address 10.1.1.1 255.255.255.252
9 R1(config-if)# clock rate 128000
10 R1(config-if)# no shutdown
11

```

Configuring Router-On-Stick for vlan routing:

```

1 R1(config)# interface fastEthernet 0/0
2 R1(config-if)# no shutdown
3 R1(config)# interface fastEthernet 0/0.10
4 R1(config-subif)# encapsulation dot1q 10
5 R1(config-subif)# ip address 192.168.10.1 255.255.255.0
6 R1(config-subif)# interface fastEthernet 0/0.20
7 R1(config-subif)# encapsulation dot1q 20
8 R1(config-subif)# ip address 192.168.20.1 255.255.255.0
9

```

Static route:

- Using next hop:

```
1 R1(config)# ip route 10.1.2.0 255.255.255.0 10.1.128.1
```

- Using exit interface:

```
1 R1(config)# ip route 10.1.2.0 255.255.255.0 Serial 0/0
```

```
2 *Note: Exit interface can be used in point-to-point serial links.
```

Default Route:

```
1 R1(config)# ip route 0.0.0.0 0.0.0.0 199.1.1.1
```

RIPv2 Configuration:

```

1 R1(config)# router rip
2 R1(config-router)# version 2
3 R1(config-router)# network 10.0.0.0      ! written as an original class A
4 R1(config-router)# no auto-summary
5 R1(config-router)# passive-interface serial 0/0

```

RIPv2 Verification:

- Shows information about the running routing protocol process:

```
1 R1# show ip protocols
```

- Shows the entire routing table:

```
1 R1# show ip route
```

- Shows routes learned via RIP only:

```
1 R1# show ip route rip
```

- Shows detailed information about the route to the specified destination network:

```
1 R1# show ip route 10.1.1.1
```

OSPF Configuration:

- Enter OSPF router configuration mode:

```
1 R1(config)# router ospf 10      ! 10 = process ID
```

- Configure one or more network commands to identify which interfaces will run OSPF:

```

1 R1(config-router)# network 10.0.0.0 0.255.255.255 area 0
2 R1(config-router)# network 172.16.8.0 0.0.7.255 area 0
3 R1(config-router)# network 192.168.1.254 0.0.0.0 area 1

```

- Configure router ID either (Optional):

Using router-id ospf subcommand:

```
1 R1(config-router)# router-id 1.1.1.1
```

Configuring an IP address on a loopback interface:

```
1 R1(config)# interface loopback 0
2 R1(config-if)# ip address 1.1.1.1 255.255.255.255
```

- Change Hello and Dead intervals per interface (Optional):

```
1 R1(config-if)# ip ospf hello-interval 2
2 R1(config-if)# ip ospf dead-interval 6
```

- Impact routing choices by tuning interface cost using one of the following ways (Optional):

Changing interface cost:

```
1 R1(config-if)# ip ospf cost 55
```

Changing interface bandwidth:

```
1 R1(config-if)# bandwidth 128      ! in Kbps
```

Changing the reference bandwidth that used by OSPF to calculate the cost:

```
1 R1(config-router)# auto-cost reference-bandwidth 1000      ! in Mbps
```

- Disabling OSPF on a certain interface (Optional):

```
1 R1(config-router)# passive-interface serial 0/0
```

- Configuring OSPF authentication (Optional):

Type 0 authentication (none):

```
1 R1(config-if)# ip ospf authentication null
```

Type 1 authentication (clear text):

```
1 R1(config-if)# ip ospf authentication
2 R1(config-if)# ip ospf authentication-key cisco
```

Type 2 authentication (md5):

```
1 R1(config-if)# ip ospf authentication message-digest
2 R1(config-if)# ip ospf message-digest-key 1 md5 cisco
```

- Configure maximum equal-cost paths (Optional):

```
1 R1(config-router)# maximum paths 6
```

OSPF verification:

- Shows information about the running routing protocol process:

```
1 R1# show ip protocols
```

- Shows the entire routing table:

```
1 R1# show ip route
```

- Shows routes learned via OSPF only:

```
1 R1# show ip route ospf
```

- Shows all neighboring routers along with their respective adjacency state:

```
1 R1# show ip ospf neighbors
```

- Shows all the information contained in the LSDB:

```
1 R1# show ip ospf database
```

- Shows detailed information about OSPF running on a specific interface:

```
1 R1# show ip ospf interfaces serial 0/0
```

EIGRP Configuration:

- Enter EIGRP configuration mode and define AS number:

```
1 R1(config)# router eigrp 121      ! 121 = AS number
```

- Configure one or more network commands to enable EIGRP on the specified interfaces:

```
1 R1(config-router)# network 10.0.0.0
2 R1(config-router)# network 172.16.0.0 0.0.3.255
3 R1(config-router)# network 192.168.1.1 0.0.0.0
4 R1(config-router)# network 0.0.0.0 255.255.255.255
```

- Disable auto summarization (Optional):

```
1 R1(config-router)# no auto-summary
```

- Disable EIGRP on a specific interface (Optional):

```
1 R1(config-router)# passive-interface serial 0/0
```

- Configure load balancing parameters (Optional):

```
1 R1(config-router)# maximum-paths 6
2 R1(config-router)# variance 4
```

- Change interface Hello and Hold timers (Optional):

```
1 R1(config-if)# ip hello-interval eigrp 121 3
2 R1(config-if)# ip hold-time eigrp 121 10
```

- Impacting metric calculations by tuning BW and delay of the interface (Optional):

```
1 R1(config-if)# bandwidth 265      ! in Kbps)
2 R1(config-if)# delay 120          ! tens of microseconds
```

EIGRP Authentication:

The *key-string* value and the mode must be the same on both routers. Lifetime options of the keys requires the clock of the routers to be set correctly, better use NTP, or it can cause problems

- Create an authentication key chain as follows:

Create a key chain and give it a name:

```
1 R1(config)# key chain MY_KEYS
```

Create one or more keys giving them numbers:

```
1 R1(config-keychain)# key 1
```

Define the key value:

```
1 R1(config-keychain-key)# key-string1stKEY
```

Define the life time of the keys (optional):

```
1 R1(config-keychain-key)# send-lifetime [start time] [end time]
2 R1(config-keychain-key)# accept-lifetime [start time] [end time]
```

- Enable md5 authentication mode for EIGRP on the interface:

```
1 R1(config-if)# ip authentication mode eigrp121 md5
```

- Refer to the correct key chain to be used on the interface:

```
1 R1(config-if)# ip authentication key-chain eigrp121 MY_KEYS
```

EIGRP Verification:

- Shows routes learned via EIGRP only:

```
1 R1# show ip route eigrp
```

- Shows EIGRP neighbors and status:

```
1 R1# show ip eigrp neighbors
```

- Shows EIGRP topology table, including successor and feasible successor:

```
1 R1# show ip eigrp topology
```

- Shows interfaces that run EIGRP:

```
1 R1# show ip eigrp interfaces
```

- Lists statistics on numbers of EIGRP messages sent and received by the router:

```
1 R1# show ip eigrp traffic
```

Access Control Lists:

Standard ACL: 1 – 99 and 1300 – 1999

- Use a remark to describe the ACL (Optional):

```
1 R1(config)# access-list 1 remark ACL TO DENY ACCESS FROM SALES VLAN
```

- Create the ACL, keeping the following in mind:

- ACL uses first-match logic.
- There is an implicit deny any at the end of the ACL.

```
1 R1(config)# access-list 2 deny 192.168.1.77
```

```
2 R1(config)# access-list 2 deny 192.168.1.64 0.0.0.31
```

```
3 R1(config)# access-list 2 permit 10.1.0.0 0.0.255.255
```

```
4 R1(config)# access-list 2 deny 10.0.0.0 0.255.255.255
```

```
5 R1(config)# access-list 2 permit any
```

- Enable the ACL on the chosen router interface in the correct direction (in or out):

```
1 R1(config-if)# ip access-group 2 out
```

- Using standard ACL to limit telnet and SSH access to a router:

Create the ACL that defines the permitted telnet clients:

```
1 R1(config)# access-list 99 remark ALLOWED TELNET CLIENTS
```

```
2 R1(config)# access-list 99 permit 192.168.1.128 0.0.0.15
```

Apply the ACL inbound the vty lines

```
1 R1(config)# line vty 0 4
```

```
2 R1(config-line)# access-class 99 in
```

Extended ACL: 100 – 199 and 2000 – 2699

- Extended ACL should be placed as close as possible to the source of the packet.
- Extended ACL matches packets based on source & des. IP addresses, protocol, source & des. Port numbers and other criteria as well

```
1 R1(config)# access-list 101 remark MY_ACCESS_LIST
```

```
2 R1(config)# access-list 101 deny ip host 10.1.1.1 host 10.2.2.2
```

```
3 R1(config)# access-list 101 deny tcp 10.1.1.0 0.0.0.255 any eq 23
```

```
4 R1(config)# access-list 101 deny icmp 10.1.1.1 0.0.0.0 any
```

```
5 R1(config)# access-list 101 deny tcp host 10.1.1.0 host 10.0.0.1 eq 80
```

```
6 R1(config)# access-list 101 deny udp host 10.1.1.7 eq 53 any
```

```
7 R1(config)# access-list 101 permit ip any any
```

```
8 R1(config)# interface fastEthernet 0/0
```

```
9 R1(config-if)# ip access-group 101 in
```

Named ACL:

- Named ACLs use names to identify ACLs rather than numbers, and commands that permit or deny traffic are written in a sub mode called named ACL mode (nacl).
- Named ACL enables the editing of the ACL (deleting or inserting statements) by sequencing statements of the ACL.
- Named standard ACL:

```
1 R1(config)# ip access-list standard MY_STANDARD_ACL
2 R1(config-std-nacl)# permit 10.1.1.0 0.0.0.255
3 R1(config-std-nacl)# deny 10.2.2.2
4 R1(config-std-nacl)# permit any
5 R1(config)# interface fastEthernet 0/1
6 R1(config-if)# ip access-group MY_STANDARD_ACL out
```

- Named extended ACL:

```
1 R1(config)# ip access-list extended MY_EXTENDED_ACL
2 R1(config-ext-nacl)# deny icmp 10.1.1.1 0.0.0.0 any
3 R1(config-ext-nacl)# deny tcp host 10.1.1.0 host 10.0.0.1 eq 80
4 R1(config-ext-nacl)# permit ip any any
5 R1(config)# interface fastEthernet 0/1
6 R1(config-if)# ip access-group MY_EXTENDED_ACL in
```

- Editing ACL using sequence numbers:

```
R1(config)# ip access-list extended MY_EXTENDED_ACL
1 R1(config-ext-nacl)# no 20      ! Deletes the statement of sequence
2 number 20
3 R1(config)# ip access-list standard 99
4 R1(config-std-nacl)# 5 deny 1.1.1.1      ! inserts a statement with
sequence 5
```

Verifying ACLs:

- Shows all ACLs configured on a router with counters at the end of each statement:

```
1 R1# show access-lists
2 ! OR
3 R1# show ip access-list
```

- Shows only the specified ACL:

```
1 R1# show ip access-list 101
```

- Includes a reference to the ACLs enabled on that interface either in or out:

```
1 R1# show ip interface f0/0
```

DHCP Server

- Define a DHCP pool and give it a name:

```
1 R1(config)# ip dhcp pool MY_POOL
```

- Define network and mask to use in this pool and the default gateway:

```
1 R1(dhcp-config)# network 192.168.1.0 255.255.255.0
2 R1(dhcp-config)# default-router 192.168.1.1
```

- Define one or more DNS server (OPTIONAL):

```
1 R1(dhcp-config)# dns-server 213.131.65.20 8.8.8.8
```

- Confine the lease time (OPTIONAL):

```
1 R1(dhcp-config) lease 2      ! Days
```

- Define one or more scopes of excluded (reserved) addresses (OPTIONAL):

```
1 R1(config)# ip dhcp excluded-address 192.168.1.1 192.168.1.100
2 R1(config)# ip dhcp excluded-address 192.168.1.200 192.168.1.254
```

DHCP Verification and Troubleshooting:

- Shows the status of the specified pool and the leased addresses from that pool:

```
1 R1# show ip dhcp pool POOL_1
```

- Shows all the leased ip addresses from all configured DHCP pools:

```
1 R1# show ip dhcp binding
```

- Shows any conflicts that occurred:

```
1 R1# show ip dhcp conflict
```

PPP Configuration:

```
1 R1(config)# interface serial 0/0
```

```
2 R1(config-if)# encapsulation ppp
```

PPP Authentication:

CHAP:

- Configure the hostname:

```
1 R1(config)# hostname ALPHA
```

- Configure the name of the other end router and the shared password:

! The password used is shared password, that means it must be the same on both routers

```
1 ALPHA(config)# username BETA password XYZ
```

- Enable CHAP authentication on the interface:

```
1 ALPHA(config)# interface serial 0/0
```

```
2 ALPHA(config-if)# ppp authentication chap
```

PAP:

- Configure the hostname:

```
1 R1(config)# hostname ALPHA
```

- Configure the name of the other end router and the shared password:

```
1 ALPHA(config)# username BETA password XYZ
```

- Enable PAP authentication on the interface and define the username and password to be sent by PAP:

```
1 ALPHA(config)# interface serial 0/0
```

```
2 ALPHA(config-if)# ppp authentication pap
```

```
3 ALPHA(config-if)# ppp pap sent-username ALPHA password XYZ
```

PPP Verification and troubleshoot:

- Shows the encapsulation type and the control protocols of PPP:

```
1 R1# show interface s0/0
```

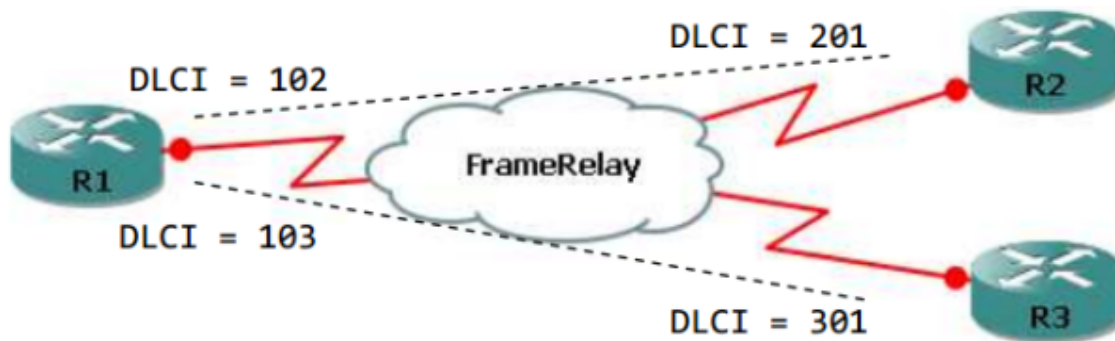
- Useful for viewing the configuration of usernames and passwords used to authenticate PPP:

```
1 R1# show running-config
```

- Displays the authentication process of PPP in real time:

```
1 R1# debug ppp authentication
```

Frame Relay:



Multipoint (one subnet)

- Give the interface an ip address and enable Frame Relay encapsulation:

```
1 R1(config)# interface serial 0/0
2 R1(config-if)# ip address 1.1.1.1 255.255.255.0
3 R1(config-if)# encapsulation frame-relay (ietf)
```

- Configure LMI signaling type: (Optional as discussed with ISP):

```
1 R1(config-if)# frame-relay lmi-type ansi      ! options: ansi, cisco,
q933a
```

- Configure Frame Relay mapping:

```
1
2 R1(config-if)# frame-relay map ip 1.1.1.2 102 broadcast (ietf)
3 R1(config-if)# frame-relay map ip 1.1.1.3 103 broadcast
4 R2(config)# interface serial 0/0
5 R2(config-if)# ip address 1.1.1.2 255.255.255.0
6 R2(config-if)# encapsulation frame-relay
7 R2(config-if)# frame-relay map ip 1.1.1.1 201 broadcast
8 R2(config-if)# frame-relay map ip 1.1.1.3 201 broadcast
9 R3(config)# interface serial 0/0
10 R3(config-if)# ip address 1.1.1.3 255.255.255.0
11 R3(config-if)# encapsulation frame-relay
12 R3(config-if)# frame-relay map ip 1.1.1.1 301 broadcast
13 R3(config-if)# frame-relay map ip 1.1.1.2 301 broadcast
14
```

Point-to-point (different subnets; one subnet per subinterface)

- Enable Frame Relay encapsulation:

```
1 R1(config)# interface serial 0/0
2 R1(config-if)# encapsulation frame-relay
```

- Give an ip address to a subinterface and configure its DLCI:

```
1
2 R1(config)# interface serial 0/0.102 point-to-point
3 R1(config-subif)# ip address 1.1.1.1 255.255.255.0
4 R1(config-subif)# frame-relay interface-dlci 102
5 R1(config)# interface serial 0/0.103 point-to-point
6 R1(config-subif)# ip address 2.2.2.1 255.255.255.0
7 R1(config-subif)# frame-relay interface-dlci 103
8 R2(config)# interface serial 0/0
9 R2(config-if)# encapsulation frame-relay
10 R2(config)# interface serial 0/0.201 point-to-point
11 R2(config-subif)# ip address 1.1.1.2 255.255.255.0
12 R2(config-subif)# frame-relay interface-dlci 201
13 R3(config)# interface serial 0/0
14 R3(config-if)# encapsulation frame-relay
15 R3(config)# interface serial 0/0.301 point-to-point
16 R3(config-subif)# ip address 2.2.2.2 255.255.255.0
17 R3(config-subif)# frame-relay interface-dlci 301
18
```


Frame Relay Verification and troubleshoot:

- Shows the encapsulation type:

```
1 R1# show interfaces serial 0/0
```

- Lists PVC status information:

```
1 R1# show frame-relay pvc
```

- Lists DLCI to IP mapping:

```
1 R1# show frame-relay map
```

- Lists LMI status information:

```
1 R1# show frame-relay lmi
```

- Displays the content of LMI messages:

```
1 R1# debug frame-relay lmi
```

- Lists messages about certain Frame Relay events, including Inverse ARP messages:

```
1 R1# debug frame-relay events
```

Network Address Translation (NAT):

Static NAT:

- Define the outside and inside interfaces:

```
1 R1(config)# interface serial 0/0
2 R1(config-if)# ip nat outside
3 R1(config)# interface FastEthernet 1/1
4 R1(config-if)# ip nat inside
```

- Configure static NAT statement:

```
1 R1(config)# ip nat inside source static 192.168.1.10 200.1.1.1
```

Dynamic NAT:

- Define the outside and inside interfaces

- Create an ACL that determines the IP addresses that are allowed to be translated:

```
1 R1(config)# access-list 3 permit 192.168.1.0 0.0.0.255
```

- Create a pool of public IP addresses:

```
1 R1(config)# ip nat pool PUB 200.1.1.1 200.1.1.6 netmask 255.255.255.248
```

- Configure NAT statement:

```
1 R1(config)# ip nat inside source list 3 pool PUB
```

```
2 NAT Overload (PAT):
```

```
3 <ul>
```

```
4 <li>The same as dynamic NAT with the use of the overload keyword at the
```

```
4 end of NAT statement:</li>
```

```
5 </ul>
```

```
6 <pre>
```

```
7
```

```
8 R1(config)# ip nat inside source list 3 pool PUB overload
```

NAT verification and troubleshoot:

- Useful in viewing the configuration of NAT pool and the inside and outside interfaces:

```
1 R1# show running-config
```

- Displays access lists, including the one used for NAT:

```
1 R1# show access-lists
```

- Shows counters for packets and NAT table entries, as well as basic configuration information:

```
1 R1# show ip nat statistics
```

- Displays the NAT table:

```
1 R1# show ip nat translations
```

- Clears all the dynamic entries in the NAT table:

```
1 R1# clear ip nat translations *
```

- Issues a log message describing each packet whose ip address is translated with NAT.

```
1 R1# debug ip nat
```